

For this project, I developed both a logistic linear regression model and a gradient boosted tree model, and evaluated each of their predictive power in terms of log loss, and selected the superior model to evaluate the shot and make probability predictions from the testing dataset. Before creating either model I wanted to evaluate the relevance of each of the parameters present in the training dataset. After splitting the `closestdefapproach` string column into 4 numerical columns for easier handling, I evaluated each variable individually by measuring their isolated predictive power on shot outcome using McFadden's pseudo r-squared. Unsurprisingly, `shottype` and `distance` were the most effective predictors, with `three` and `locationx` also performing well due to their high collinearity with the first two. Next, I wanted a way to adjust the `shooter_id` and other id variables so that they handled players with a low number of shot attempts correctly. Players with few shot attempts could have an artificially inflated make rate that would cause the model to overestimate the make probability for these players. With the help of Claude, I smoothed these instances out by implementing pseudo shots for these players at the global mean make rate. While imperfect due to players with few shots presumably being worse shooters than the global average, it still defends against unreasonably high averages on low shot players.

After these modifications I fit the most relevant variables into a basic binomial logistic regression model with no interaction terms while omitting certain variables that I deemed to be highly collinear with another variable (ex: `contested/num_contesters` or `shottype/three`). With the help of Claude, I evaluate the base model in terms of log loss since that is performance criteria measured upon submission. The base model performed adequately, with a mean CV log-loss of 0.6423871 after four folds and an in-sample log-loss of 0.6423545. Importantly, the in and out of sample performances are nearly identical, indicating a lack of overfitting as expected for a basic model.

To develop the model further I assessed the following interactions in terms of log loss change when applied to the base model alone: `shottype:distance` (distance affects shot types differently), `shottype:closestdefdist` (coverage punishes jumpers more than layups and dunks), `num_contesters:distance` (longer shots are more difficult with contesters), `shottype:shooterspeed` (speed effects layups and dunks significantly less than jumpers), `gamestate:shotclock`, (pressure compounds with close game and less time), `three:closestdefdist` (threes are much harder in close coverage) `shottype:shotclock` (jumpers with low time are typically lower percentage) and `dribblesbefore:shotclock`. Only `shottype:closestdefdist` (improved log-loss by 0.00400) `three:closestdefdist` (improved log-loss by 0.00319) `shottype:distance` (improved log-loss by 0.00306) and `num_contesters:distance` (improved log-loss by 0.00268) had a meaningful impact on the model's predictive accuracy, so only those were added.

The final binomial logistic model including interaction terms had an in-sample log-loss of 0.633909 and a mean CV log-loss of 0.6340068, a marked improvement over the basic model while still avoiding overfitting. However, rather than continue to guess at higher order interaction terms that could possibly improve performance, I felt that using a tree model would capture some of these untested interactions automatically and possibly outperform the logistic model.

Using xgboost, I created a gradient boosted tree model that creates a maximum of 500 decision trees, which are designed to split at levels that minimize mistakes in shot prediction. Each tree has a maximum depth of 6 splits, and sees a random 80% of the data, both of which guard against overfitting. As the model trains itself, it measures its progress on log loss, and stops generating trees after not seeing any log-loss improvement in the previous 20 generated trees. The trees are made only using the relevant variables incorporated in the base model.

The tree model performed significantly better than the logistic model, with a mean CV log-loss of 0.6287705. I elected to use the tree model for my submission due to its superior performance; all outputs in submission.csv are made using the tree model. Given more time, I would possibly reevaluate the tree model to include more variables from the initial dataset, since multicollinearity is less of an issue if not using a logistic regression. Additionally, the shooter_id smoothing process could be improved or become unnecessary with access to more data on shooters with few shot attempts. This variable was also used in a limited fashion by handling a player's shot making average as one figure across all types of shots. Using shooter id to calculate players average for each type of shot would improve the predictive power of that variable, since undoubtedly players have meaningful differences in shot making ability depending on the type of shot (like Rudy Gobert taking a three). Moreover, the inclusion of a binary variable for whether or not a player was fouled on their shot attempt would significantly improve the predictive power of the model, as shots made after the foul are significantly more difficult and lower percentage. Other variables not featured in the set that may improve prediction could include ball release height, and time with ball before the shot.